



CHAPTER

Pretreatment and Filtration Skid

Disc filters, ultrafiltration unit and pretreatment piping

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8.1 Overview and function

The pretreatment container conditions raw feed water before it reaches the SonixED™ stacks: disc filtration removes coarse suspended solids, and an ultrafiltration (UF) stage removes fine particulates and reduces fouling potential. Filtered water passes to the treatment container (chapter 3); rejected solids and backwash water are routed to the drain tank.



Figure 8.1. Suggested figure: photograph of the assembled pretreatment skid inside the container, with the disc filters, UF unit and main piping runs labeled.

8.2 Key specifications

- **Ultrafiltration:** Akualys/AKKIM APM 80 hollow-fiber UF module(s), 2" Victaulic connections. Two modules are confirmed ordered for HydroVolta (160 m² total membrane area). Membrane: modified PVDF hollow fiber, outside-to-inside flow, fiber OD/ID 1.4/0.8 mm, nominal MWCO ≤150,000 Da (≈30 nm pore size), 80 m² active area per module.
- **UF feedwater limits:** temperature ≤40 °C (typ. 25 °C); turbidity ≤250 NTU (typ. 50 NTU); TSS ≤80 mg/l; TOC ≤30 mg/l; pH 6–9; instantaneous free chlorine ≤150 mg/l.
- **UF operation:** dead-end/crossflow, filtrate flux 45–180 l/m²h at 25 °C, flow capacity 3.60–14.40 m³/h per module, feed inlet pressure 2–3 bar (instantaneous max. 5 bar), transmembrane pressure (TMP) 0.4–2 bar, filtrate SDI ≤2.5, filtrate turbidity ≤0.2 NTU.
- **Disc filtration:** AZUD Helix Automatic self-cleaning disc filters (FT-series, automatic backwash), 100-micron mesh discs; the documented BASF pilot P&ID additionally references disc-filter positions DF100/DF50/DF20 by micron rating — TODO confirm which AZUD Helix model number(s) map to the standard SalinBloc™ pretreatment train.
- Feed inlet: 2½" (M); filtered-water outlet: 2" (M); drain: 3" (M).
- Feed pump (pilot configuration): submersible, 4.5 m³/h at 6 bar.

TODO: confirm whether the AKKIM/AZUD models above are the standard SalinBloc™ bill-of-materials or pilot-specific; a finalized pretreatment BOM was not found. Several other filtration technologies (hydrocyclone separator, automatic screen filters, high-efficiency sand filters, microfilters) exist as vendor catalogs in HydroVolta's component library but are not confirmed as installed on any SonixED™ pretreatment train.

8.3 Operation

No dedicated pretreatment-specific operating procedure beyond the general startup/shutdown described in chapter 6 was found, but the UF module manufacturer documents the following normal operating cycle, which is the closest available reference for the standard SalinBloc™ pretreatment skid:

1. **Production:** a brief forward-flush step, then filtration (dead-end or crossflow) until the trigger for backwash is reached.
2. **Backwash:** drain, then air-scour (max. 20 Nm³/h at 600–800 mbar) to loosen the fouling layer, then backwash bottom-to-top and top-to-bottom (backwash pressure max. 2.5 bar) to flush it out.

TODO: confirm the actual backwash trigger used on SalinBloc™ (pressure differential vs. timer) and the cycle duration/frequency; the manufacturer spec above describes the module's capability, not HydroVolta's configured setpoint.

8.4 Maintenance

The UF manufacturer documents a Chemically Enhanced Backwash (CEB) / CIP cycle, triggered when TMP rises ≥ 0.9 bar above its initial value:

1. Run an extended backwash cycle (as in *Operation* above), then soak the module in cleaning chemical for part of the cycle.
2. CEB chemical concentrations: NaOCl 1000 mg/l, NaOH 500 mg/l, HCl 1000 mg/l, or citric acid 1–2% — select per the fouling type (organic/biological vs. scale) per chapter 12.
3. Total CIP duration ≈ 2 hours, at a cleaning flow of 1–2 m³/h per module.

For extended shutdowns (chapter 20), the manufacturer specifies a glycerol/sodium-metabisulfite preservation soak, and requires a minimum 4-hour flush with ≥ 14 m³ of feed water before restart. Before returning a preserved or newly installed module to service, disinfect with 100 ppm NaOCl circulated for 10 minutes, held for 1 hour, then drained until the filtrate runs chlorine-free. TODO: no document gives a routine cleaning-interval schedule (as opposed to the TMP-triggered CEB above) — see chapter 7 for the general log template.

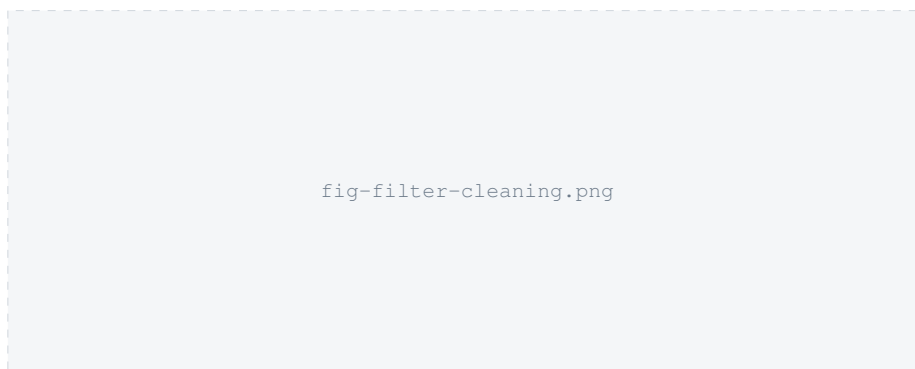


Figure 8.2. Suggested figure: step-by-step photos of the disc-filter backwash and UF CEB/CIP procedure above, once photographed on an actual unit.

8.5 Troubleshooting

- **Low filtered-water flow / high pressure drop:** check pump duty point against the 4.5 m³/h @ 6 bar rating and pipe sizing (60 mm feed line, 40 mm filtered-water line) before suspecting the filters themselves.
- **Rising transmembrane pressure (TMP):** if TMP has risen ≥ 0.9 bar above its initial value, run the CEB/CIP cycle above rather than waiting for a scheduled cleaning.
- **Poor downstream water quality:** verify feed water against the operating envelope in chapter 10 (turbidity, iron, SDI) and against the UF feedwater limits above — an out-of-envelope feed is a likely root cause on either stage.

TODO: no documented fault table (e.g. specific alarm codes) for the disc filters or UF unit was found beyond the operating-envelope figures above.